

# MKID Energy Resolution and Its Impact on Fourier Transform Spectroscopy Performance

Irmak Aytekin  
CTA200H

May 17, 2026

## 1 Introduction

Fourier Transform Spectrographs (FTS) achieve high spectral resolution by scanning a Michelson interferometer through many mirror positions, encoding wavelength information in the resulting interferogram. However, when observing broadband light (e.g., the full visible spectrum), different spectral orders overlap and cannot be distinguished, a fundamental limitation known as *order confusion*.

Microwave Kinetic Inductance Detectors (MKIDs) are superconducting photon-counting detectors that provide an independent energy measurement for each incoming photon. While this energy resolution ( $R_E \sim 10 - 100$ ) is modest compared to the FTS spectral resolution ( $R \sim 1000 - 10000$ ), it is sufficient to split incoming light into distinct wavelength bands, eliminating order overlap within each band and unlocking the full resolving power of the interferometer.

This report demonstrates the impact of MKID energy resolution on the usable spectral resolution of a combined MKID+FTS instrument.

## 2 Instrument Parameters

We consider an ideal Michelson interferometer with the following specifications:

- Number of mirror steps:  $N = 1000$
- Step size:  $\delta = 1 \mu\text{m}$
- Maximum optical path difference (OPD):  $\text{OPD}_{\text{max}} = 2 \times N \times \delta = 2 \text{ mm}$
- Wavelength range:  $\lambda = 400 - 700 \text{ nm}$  (visible)

The factor of 2 in the OPD arises because light travels to the moving mirror and back.

## 3 Part 1: Spectral Resolution Without MKID

The fundamental spectral resolution of an FTS is given by:

$$\delta\lambda = \frac{\lambda^2}{\text{OPD}_{\text{max}}} \tag{1}$$

The resolving power is:

$$R = \frac{\lambda}{\delta\lambda} = \frac{\text{OPD}_{\text{max}}}{\lambda} \quad (2)$$

For our instrument, this yields:

- At  $\lambda = 400 \text{ nm}$ :  $\delta\lambda = 0.080 \text{ nm}$ ,  $R = 5000$
- At  $\lambda = 550 \text{ nm}$ :  $\delta\lambda = 0.151 \text{ nm}$ ,  $R = 3636$
- At  $\lambda = 700 \text{ nm}$ :  $\delta\lambda = 0.245 \text{ nm}$ ,  $R = 2857$

Figure 1 shows how spectral resolution and resolving power vary across the visible range. While the theoretical resolution is excellent, **this resolution is not usable across the full visible band** due to order overlap: a 400 nm photon and an 800 nm photon (one order higher) produce identical interferograms and cannot be distinguished.

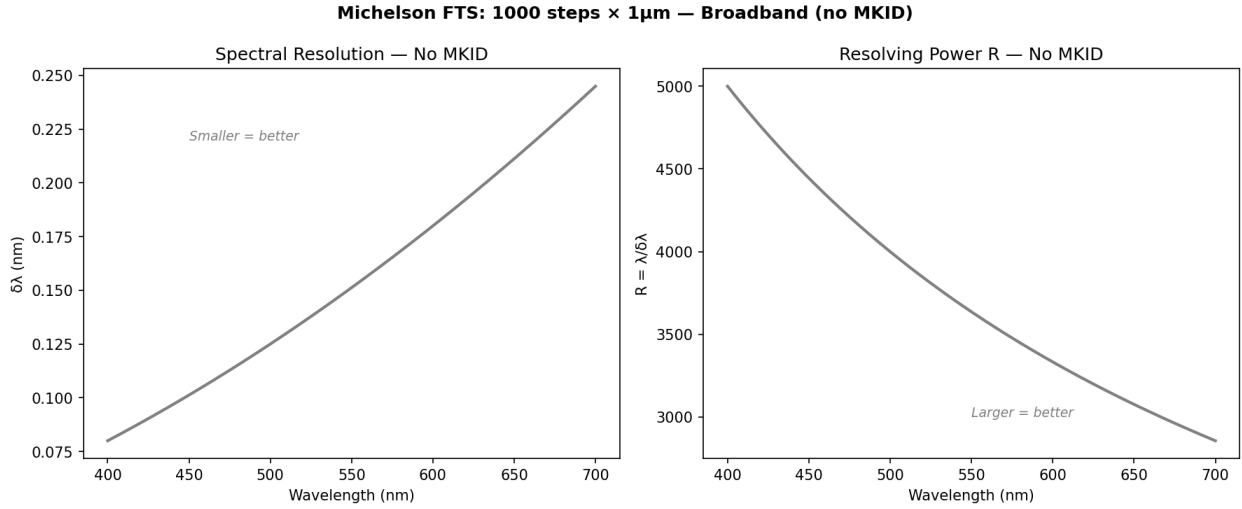


Figure 1: Spectral resolution and resolving power of the Michelson FTS without MKID energy discrimination. While the theoretical resolution is high, order confusion limits its practical usability across the full visible band.

## 4 Part 2: Adding MKID Energy Resolution

An MKID with energy resolution  $R_E = 25$  can resolve photon energies to  $\Delta E/E = 1/R_E = 4\%$ . At a given wavelength  $\lambda$ , this corresponds to a wavelength band width of:

$$\Delta\lambda_{\text{band}} = \frac{\lambda}{R_E} \quad (3)$$

For  $R_E = 25$  across the visible range:

- At  $\lambda = 400 \text{ nm}$ : band width = 16.0 nm
- At  $\lambda = 550 \text{ nm}$ : band width = 22.0 nm
- At  $\lambda = 700 \text{ nm}$ : band width = 28.0 nm

This splits the 400 – 700 nm range into approximately **13-14 distinct bands**. Within each band, order confusion is eliminated, and the full FTS resolution ( $\delta\lambda \sim 0.08 - 0.25$  nm) becomes usable.

Figure 2 shows the band structure (left) and the dramatic improvement in usable resolving power (right). The MKID band width is  $\sim 200\times$  larger than the FTS resolution, meaning the MKID only needs to provide coarse wavelength sorting, the FTS then delivers fine spectral resolution within each band.

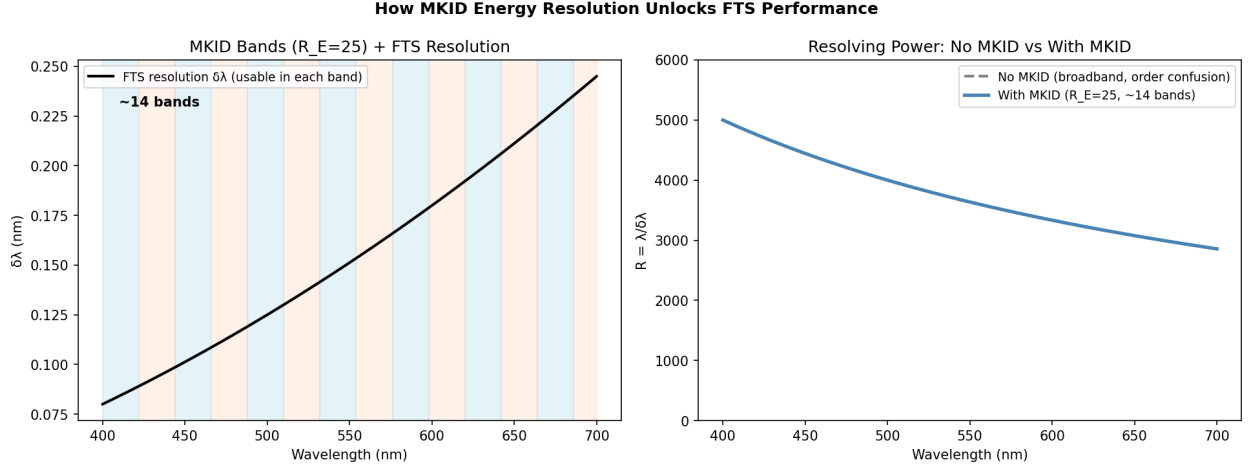


Figure 2: Left: MKID splits visible light into  $\sim 14$  bands (shaded regions), each containing the full FTS resolution. Right: Comparison of usable resolving power with and without MKID.

## 5 Part 3: Impact of MKID Energy Resolution

Figure 4 shows how varying the MKID energy resolution  $R_E$  affects instrument performance. Key observations:

- $R_E = 5$ : Only  $\sim 3$  bands across visible. Still much better than no MKID, but limited spectral coverage per band.
- $R_E = 10$ :  $\sim 7$  bands. Resolving power approaches the full FTS limit.
- $R_E = 25$ :  $\sim 14$  bands. Excellent performance, narrow bands eliminate order confusion while maintaining high resolution.
- $R_E = 50 - 100$ :  $\sim 27 - 54$  bands. Marginal improvement beyond  $R_E = 25$  for this particular FTS configuration.

**Conclusion:** Even a modest MKID energy resolution ( $R_E \sim 10 - 25$ ) is sufficient to unlock the full spectral resolution of the interferometer. Higher  $R_E$  provides finer band splitting but offers diminishing returns once order confusion is adequately resolved.

## 6 Discussion

This analysis demonstrates that **MKID energy resolution and FTS spectral resolution are complementary technologies**. The MKID's coarse energy discrimination ( $R_E \sim 10 - 50$ )

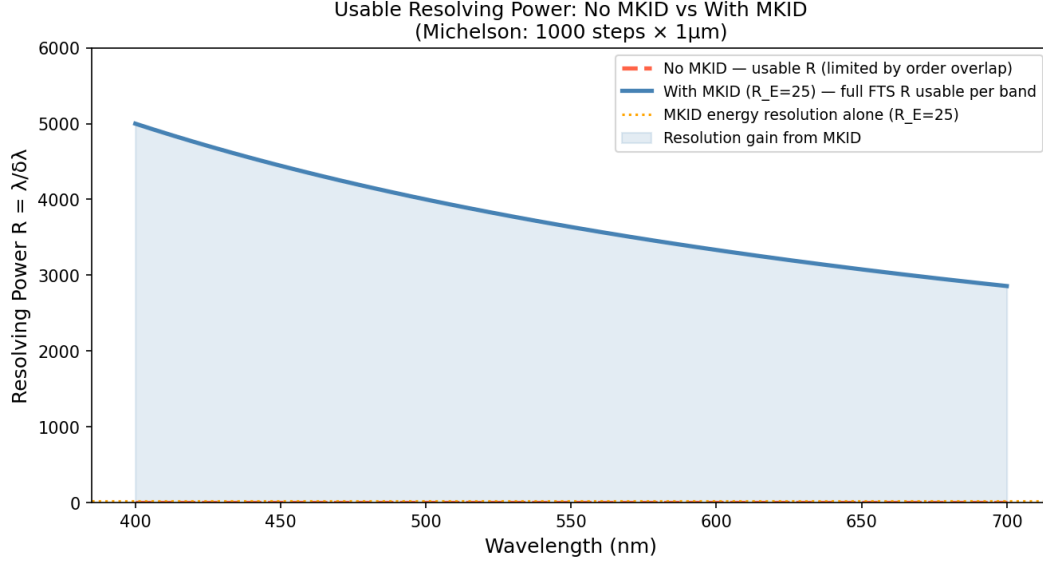


Figure 3: Usable resolving power comparison. Without MKID, order confusion limits usable  $R$  to  $\sim 400 - 700$ . With MKID ( $R_E = 25$ ), the full FTS resolution ( $R \sim 3000 - 5000$ ) becomes accessible. The shaded region shows the resolution gain enabled by the MKID.

eliminates the order overlap problem, allowing the interferometer to deliver its full resolution ( $R \sim 3000 - 5000$ ) across a broad wavelength range.

For the lab instrument under development, an MKID with  $R_E \sim 20 - 30$  is sufficient to achieve excellent performance across the visible band when paired with a 2 mm OPD scan. This capability is essential for the data reduction pipeline, as photon events must be sorted into wavelength bands before interferogram construction.

Future work should consider the practical effects of MKID energy resolution degradation at high count rates, temperature drift, and pixel-to-pixel variation in  $R_E$  across the detector array.

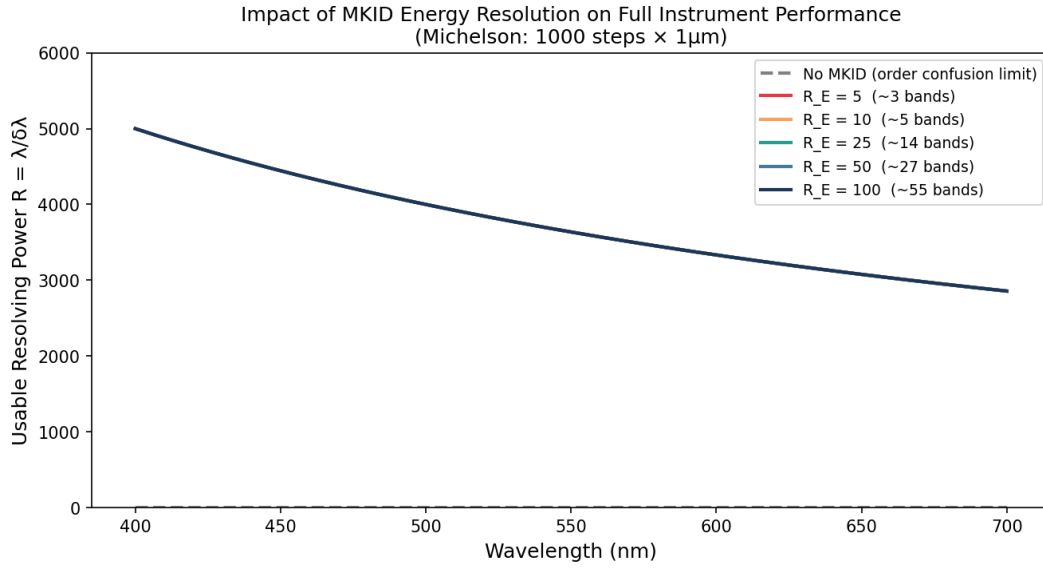


Figure 4: Effect of varying MKID energy resolution on usable resolving power. The gray dashed line shows the order-confusion-limited case (no MKID). Higher  $R_E$  enables more bands and better performance, but returns diminish beyond  $R_E \sim 25$  for this instrument.